

NETWORKWALKS

WEEK 4 • AUTHORIZED SECURITY ASSESSMENT

PENETRATION TESTING REPORT

MEDIROZA GENERAL HOSPITAL

<https://medirozahospital.com>

Prepared by	Adewuyi Wakeel (Pen Tester)
Course / Engagement	Cybersecurity Int. B083A — Waqas Karim, CCIE
Assessment Type	Authorized Black-Box Penetration Test
Client / Target	Mediroza General Hospital
Authorization	Written permission granted for the controlled training environment
Assessment Date	30th September 2026
Classification	CONFIDENTIAL

01 — Engagement Overview

Authorization & Liability

This report covers cybersecurity testing carried out solely within the sanctioned Mediroza General Hospital training environment. The engagement took the form of a controlled black-box assessment targeting the web application. Testing was confined to the approved target; denial-of-service attacks, social engineering, and testing of unrelated systems were excluded.

The project serves an educational purpose: identifying security flaws, assessing their potential impact, and outlining practical remediation. The techniques described should not be used against systems without explicit authorization from the system owner.

Executive Summary

The Week 4 engagement followed a realistic penetration-testing workflow from reconnaissance through impact analysis and reporting. The supplied evidence confirms the public-facing Mediroza website, a dedicated patient login portal, sensitive paths disclosed through robots.txt, and a patient-report area containing three encrypted pathology reports.

The broader workflow included controlled authentication testing, password-strength assessment of protected PDFs, investigation of legacy content, and review of potential exposure of employee or shareholder information. Patient information, recovered passwords, salary figures, and shareholder details are intentionally withheld from this report.

Scope & Methodology

Target	https://medirozahospital.com
Engagement model	Authorized educational black-box web-application assessment
Primary objectives	Identify entry points; evaluate authentication and access control; examine protected patient documents; assess document-password strength; investigate legacy resources; check sensitive organizational-data exposure; provide remediation guidance.
Rules of engagement	Approved domain only. No denial-of-service, social engineering, destructive changes, or out-of-scope systems.
Evidence policy	Only supplied screenshots are treated as visual proof; other stages are described narratively without fabricated evidence.

02 — Tools & Assessment Workflow

Tool / Technique	Purpose
Web Browser / Firefox	Manual navigation, authentication review, portal validation, and inspection of web-accessible resources.
robots.txt Review	Passive discovery of paths excluded from search-engine crawling but visible to users.
curl	Direct HTTP request/response inspection and validation of web resources.
SQL Injection Testing	Authorized assessment of whether authentication inputs are handled safely by the back-end application.
Browser Developer Tools	Inspection of page source and login-form behaviour.
Networkwalks Hash Calculator / Password Cracker	Lab tools for PDF hash extraction and dictionary-based password assessment.
Text / Database Analysis Utilities	Review of retrieved legacy backup material for sensitive-data exposure without altering original evidence.
Claude	Conversion of raw SQL data into readable tables during analysis.

Assessment Sequence

The work was arranged chronologically so that each stage answered a specific security question and demonstrated how one weakness could increase the impact of another.

1. Initial reconnaissance and application mapping
2. Patient-portal authentication review
3. Passive discovery through robots.txt
4. Patient-report area review
5. Controlled retrieval of protected PDF reports
6. PDF encryption and hash preparation
7. Authorized password-recovery assessment
8. Verification of decrypted documents
9. Legacy /old/ resource review
10. Database-backup and sensitive-information review

03 — Technical Assessment

3.1 Initial Reconnaissance & Application Mapping

The public-facing Mediroza General Hospital website was examined first. Its homepage displayed navigation items including Home, About, Doctors, Contact, Staff Login, and Patient Portal. The dedicated patient authentication section was treated as a high-value target because access to it could expose confidential medical records.

3.2 Patient Portal Authentication Surface

The Patient Portal presented a standard username/password form, a Sign-in button, and an option to reveal the typed password. The authorized scope included controlled SQL-injection testing against the login system to determine whether attacker-supplied values could alter an underlying database query. A secure implementation should treat input as data, use parameterized queries, and reject access without valid credentials.

3.3 Public Discovery of Sensitive Paths through robots.txt

The robots.txt file listed three disallowed paths — /patient/, /staff/, and /old/ — together with a sitemap link. A robots.txt entry is not an access-control mechanism; it is a request to compliant crawlers to avoid indexing a path. Sensitive directories therefore still require real authentication and authorization. The /old/ path was particularly relevant because legacy resources can contain forgotten backups, outdated files, or older application components.

3.4 Patient Report Area

The evidence described a patient-report section containing three downloadable pathology reports, each labelled with a patient's name and marked as an encrypted PDF. A note indicated that report passwords came from the laboratory. This creates two security layers: portal access control and document-level password protection.

3.5 Controlled Retrieval of Protected PDFs

The three reports were downloaded into the controlled test environment for authorized offline assessment. This confirmed that the risk extends beyond webpage visibility to acquisition of protected files. The files were treated as sensitive evidence and retained only within the project's findings area.

3.6 PDF Encryption and Hash Preparation

Each protected PDF was handled independently. The process identified the encryption type and extracted a password-verification hash using the Networkwalks Hash Calculator. The hash does not directly reveal the password; it provides a reference against which candidate passwords can be tested separately. Actual hash values are intentionally omitted.

3.7 Authorized Password-Recovery Assessment

The extracted PDF hashes were subjected to dictionary-based password testing using the Networkwalks Password Cracker. Each file was assessed individually. The purpose was to determine whether weak, reused, predictable, or common passwords could undermine document encryption. Specific recovered passwords are intentionally omitted.

3.8 Verification of Decrypted Documents

A recovered password is treated as confirmed only when it successfully unlocks its corresponding PDF in the controlled environment. Confidential medical content is not reproduced unnecessarily in the final report.

04 — Legacy Resources & Data Exposure

4.1 Legacy /old/ Resource Review

The /old/ path identified during reconnaissance was singled out for follow-up because legacy directories can contain obsolete components, archived material, or backup data that remains reachable after active use has ended. The project workflow specifically called for investigating previously gathered material for a further significant exposure on the server.

4.2 Database Backup Exposure

Database backups can contain substantially more information than normal application pages, including user records, internal identifiers, historical entries, credentials or password hashes, employee data, and ownership details. A backup reachable through the web should therefore be treated as a serious configuration issue even when the live application otherwise enforces access controls.

4.3 Employee Salary Information

The review included searching recovered database content for employee records and salary fields. Exposure of salary information represents a confidentiality issue and can increase privacy, fraud, social-engineering, and reputational risks. Actual names and figures are withheld from this report.

4.4 Shareholder and Ownership Information

The recovered material was also reviewed for shareholder-related information such as names, share allocations, contact details, or ownership records. Such information should remain within protected project evidence rather than a public-facing report. The finding is framed as sensitive corporate-data exposure associated with insufficient protection of legacy backup files.

Data-handling note: Detailed patient, employee, financial, ownership, hash, and password evidence should remain restricted to the authorized project evidence set. This report deliberately summarizes the categories of exposure rather than reproducing sensitive records.

05 — Risk Analysis

The ratings below reflect the likely security significance of the Week 4 findings. They are based on the supplied report's assessment and distinguish direct observations from later-stage impacts documented through the authorized workflow.

1	Patient authentication weakness / potential SQL injection	High	Unsafe input handling could permit authentication bypass and access to protected medical resources.
2	Protected patient reports reachable after portal access	High	Unauthorized portal access can expose confidential reports and enable offline attacks against document passwords.
3	Sensitive paths disclosed in robots.txt	Low	Reveals /patient/, /staff/, and /old/, aiding reconnaissance but not bypassing access control by itself.
4	Weak PDF password protection	High	Predictable or crackable document passwords can undermine encryption after files are obtained.
5	Legacy /old/ content exposed	High	Forgotten resources may expose backup files or outdated components outside normal application controls.
6	Database backup accessible from web content	Critical	A single exposed backup can disclose large volumes of application, employee, and corporate data.
7	Employee salary information exposure	High	Confidential HR data can create privacy, targeted-social-engineering, and reputational risks.
8	Shareholder information exposure	High	Sensitive ownership information can create commercial, privacy, and targeted-fraud risks.

06 — Recommendations & Remediation

1. Use parameterized queries

All authentication and data-access queries should use prepared statements or parameterized APIs. User input should never be concatenated directly into SQL.

2. Strengthen authentication controls

Implement secure credential verification, generic error messages, rate limiting, secure session management, and server-side authorization checks on every protected resource.

3. Do not rely on robots.txt for protection

Sensitive paths require genuine authentication and authorization. Remove unnecessary directory references and avoid exposing legacy locations publicly.

4. Remove or isolate legacy content

Delete obsolete directories from the production web root. If retention is necessary, move backups to non-web-accessible storage with restricted permissions.

5. Protect database backups

Never serve backups through the web application. Encrypt backups at rest, restrict access, apply retention policies, and monitor backup repositories.

6. Use strong document protection

Apply long, unique, randomly generated passwords to protected files and avoid predictable lab, patient, date, or dictionary-based passwords.

7. Prefer secure document delivery

Where possible, use authenticated portal access and protected server-side storage rather than long-lived encrypted documents that can be attacked offline.

8. Minimize sensitive-data exposure

Store only necessary employee and shareholder data, apply least privilege, and redact or tokenize highly sensitive fields where feasible.

9. Centralize logging and alerting

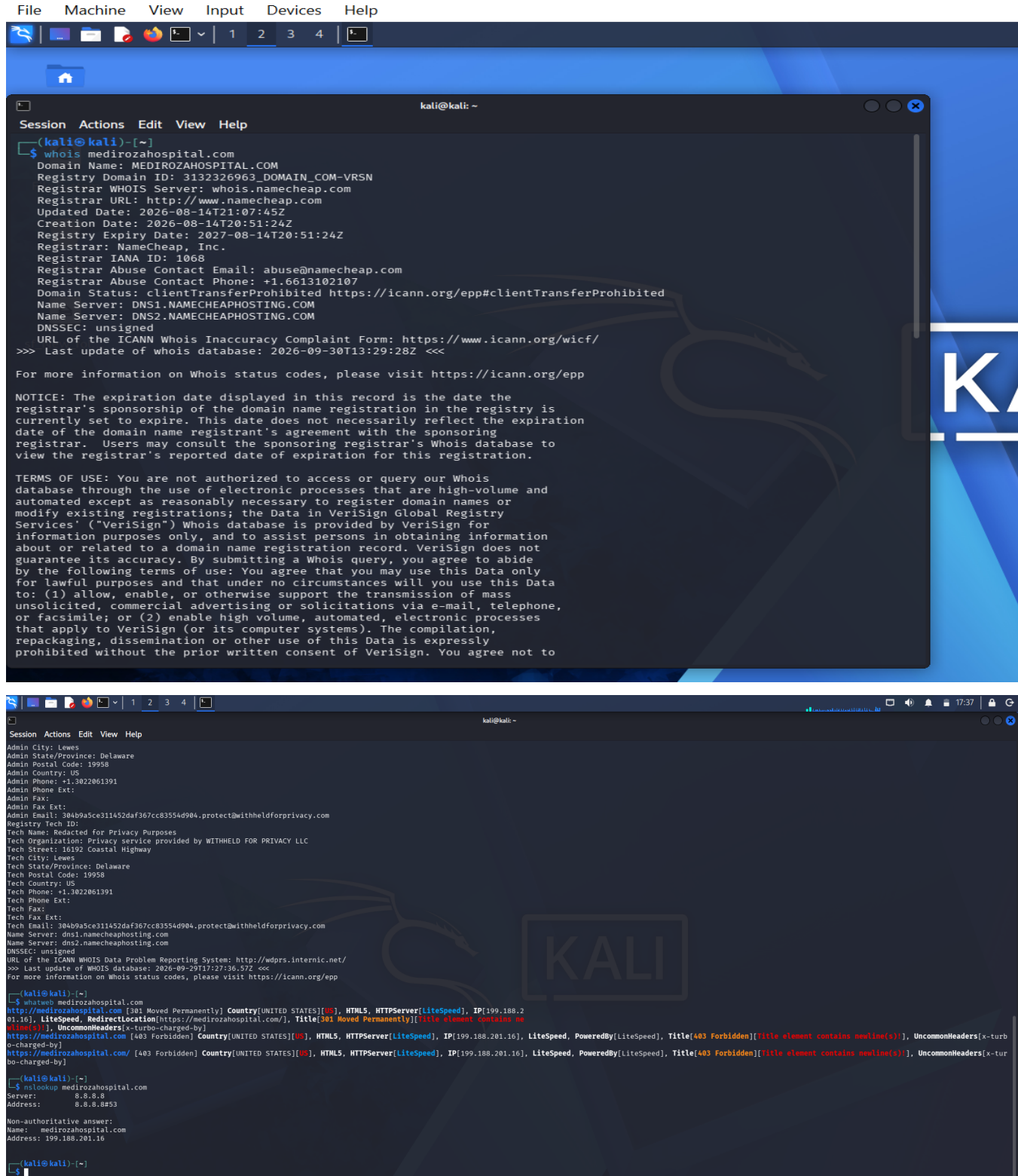
Monitor repeated authentication failures, abnormal file downloads, directory enumeration, and access to legacy or backup locations.

10. Conduct periodic security reviews

Repeat authorized penetration tests after remediation, including regression testing for authentication, access control, backup exposure, and sensitive-file handling.

07 — Evidence Register

All screenshots supplied below by the pen tester are treated as embedded visual evidence. No substitute images are created for stages without user-provided visual evidence from basic reconnaissance to analysis of significant confidentiality information.



```
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

kali@kali:~$ whois medirozahospital.com
Domain Name: MEDIROZAHOSPITAL.COM
Registry Domain ID: 3132326963_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.namecheap.com
Registrar URL: http://www.namecheap.com
Updated Date: 2026-08-14T21:07:45Z
Creation Date: 2026-08-14T20:51:24Z
Registry Expiry Date: 2027-08-14T20:51:24Z
Registrar: NameCheap, Inc.
Registrar IANA ID: 1068
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.6613102107
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: DNS1.NAMECHEAPHOSTING.COM
Name Server: DNS2.NAMECHEAPHOSTING.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-30T13:29:28Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to

kali@kali:~$ nslookup medirozahospital.com
Server: 8.8.8.8
Address: 8.8.8.853

Non-authoritative answer:
Name: medirozahospital.com
Address: 199.188.201.16

kali@kali:~$
```

[illegible]

```
kali@kali -  
[--(kali@kali)-~]  
$ dnsrecon -d medirozahospital.com  
2026-09-30T17:52:39.642370-0400 INFO Starting enumeration for domain: medirozahospital.com  
2026-09-30T17:52:39.644989-0400 INFO std: Performing General Enumeration against: medirozahospital.com...  
2026-09-30T17:52:40.332051-0400 ERROR No answer for DNSSEC query for medirozahospital.com.  
2026-09-30T17:52:40.765788-0400 INFO SOA dns1.namecheaphosting.com 156.154.132.200  
2026-09-30T17:52:40.766472-0400 INFO SOA dns1.namecheaphosting.com 2618x11:1024;200  
2026-09-30T17:52:44.367583-0400 INFO NS dns2.namecheaphosting.com 156.154.133.200  
2026-09-30T17:52:45.291269-0400 INFO Bind Version for 156.154.133.200 Nameserver  
2026-09-30T17:52:45.292266-0400 INFO NS dns2.namecheaphosting.com 2618x11:1024;200  
2026-09-30T17:52:45.293969-0400 INFO NS dns1.namecheaphosting.com 156.154.132.200  
2026-09-30T17:52:45.612944-0400 INFO Bind Version for 156.154.132.200 Nameserver  
2026-09-30T17:52:47.028390-0400 INFO MX mx-hosting-jellyfish.systems 102.255.118.20  
2026-09-30T17:52:47.029051-0400 INFO MX mx2-hosting-jellyfish.systems 102.255.118.29  
2026-09-30T17:52:47.029393-0400 INFO MX mx3-hosting-jellyfish.systems 102.255.118.13  
2026-09-30T17:52:47.681733-0400 INFO A medirozahospital.com 199.188.201.16  
2026-09-30T17:52:48.365202-0400 INFO TXT medirozahospital.com v=spf1 +a +mx +ip4:199.188.201.7 +ip4:199.188.201.38 inc  
lude:spf.web-hosting.com ~all  
2026-09-30T17:52:48.65931-0400 INFO TXT _dmarc.medirozahospital.com v=DMArc1;p=none;  
2026-09-30T17:52:49.024180-0400 INFO Enumerating SRV Records  
2026-09-30T17:52:54.734975-0400 INFO SRV _caldav._tcp.medirozahospital.com medirozahospital.com 199.188.201.16 2080  
2026-09-30T17:52:54.744951-0400 INFO SRV _caldav._tcp.medirozahospital.com medirozahospital.com 199.188.201.16 2079  
2026-09-30T17:52:54.791327-0400 INFO SRV _cardav._tcp.medirozahospital.com medirozahospital.com 199.188.201.16 2079  
2026-09-30T17:52:54.855292-0400 INFO SRV _cardavs._tcp.medirozahospital.com 199.188.201.16 2080  
2026-09-30T17:52:55.215703-0400 INFO SRV _autodiscover._tcp.medirozahospital.com cpnemaildiscovery.cpanel.net 184.9  
4.204.9 443  
2026-09-30T17:52:55.216090-0400 INFO SRV _autodiscover._tcp.medirozahospital.com cpnemaildiscovery.cpanel.net 184.9  
4.204.9 443  
2026-09-30T17:52:55.216474-0400 INFO SRV _autodiscover._tcp.medirozahospital.com cpnemaildiscovery.cpanel.net 184.9  
4.204.14 443  
2026-09-30T17:52:55.216848-0400 INFO SRV _autodiscover._tcp.medirozahospital.com cpnemaildiscovery.cpanel.net 184.9  
4.203.15 443  
2026-09-30T17:52:55.217019-0400 INFO SRV _autodiscover._tcp.medirozahospital.com cpnemaildiscovery.cpanel.net 184.9  
4.203.14 443  
2026-09-30T17:52:55.217373-0400 INFO SRV _autodiscover._tcp.medirozahospital.com cpnemaildiscovery.cpanel.net 184.9  
4.204.15 443  
2026-09-30T17:52:55.217534-0400 INFO SRV _autodiscover._tcp.medirozahospital.com cpnemaildiscovery.cpanel.net 184.9  
4.203.11 443  
2026-09-30T17:52:55.217930-0400 INFO SRV _autodiscover._tcp.medirozahospital.com cpnemaildiscovery.cpanel.net 184.9  
4.204.11 443  
2026-09-30T17:52:55.219388-0400 INFO 12 Records Found  
2026-09-30T17:52:55.219959-0400 INFO Completed enumeration for domain: medirozahospital.com  
  
[--(kali@kali)-~]  
$
```

```

kali@kali: ~
Session Actions Edit View Help
$ curl -I https://medirozahospital.com

HTTP/2 200
content-type: text/html
last-modified: Fri, 04 Sep 2026 02:03:12 GMT
accept-ranges: bytes
content-length: 5445
date: Wed, 30 Sep 2026 14:27:01 GMT
server: LiteSpeed
x-turbo-charged-by: LiteSpeed

<!-- index.html -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="author" content="Mediroza IT Department">
  <meta name="description" content="Mediroza General Hospital - trusted healthcare, modern facilities and specialist doctors serving the community.">
  <meta name="generator" content="Mediroza CMS 1.4.2">
  <title>Mediroza General Hospital | Compassionate Care, Advanced Medicine</title>
  <link rel="stylesheet" href="/styles.css">
</head>
<body>

<div class="topbar">
  <div class="container">
    <span>Emergency: +27 11 555 0100 </span> Open 24/7</span>
    <span><a href="/staff/login.php">Staff Login</a> </span>
  </div>
</div>

<header class="header">
  <div class="container">
    <a class="brand" href="/index.html">
      <span class="logo-box">
        <svg viewbox="0 0 24 24"><path d="M12 5v14M5 12h14" stroke="ffff" stroke-width="2.5" stroke-linecap="round"/></svg>
      </span>
      <span class="wordmark">Medi<span class="g">roza</span></span>
    </a>
    <nav class="nav">
      <a href="/index.html">Home</a>
      <a href="/about.html">About</a>
      <a href="/doctors.html">Doctors</a>
      <a href="/contact.html">Contact</a>
      <a class="btn-portal" href="/patient/login.php">Patient Portal</a>
    </nav>
  </div>
</header>

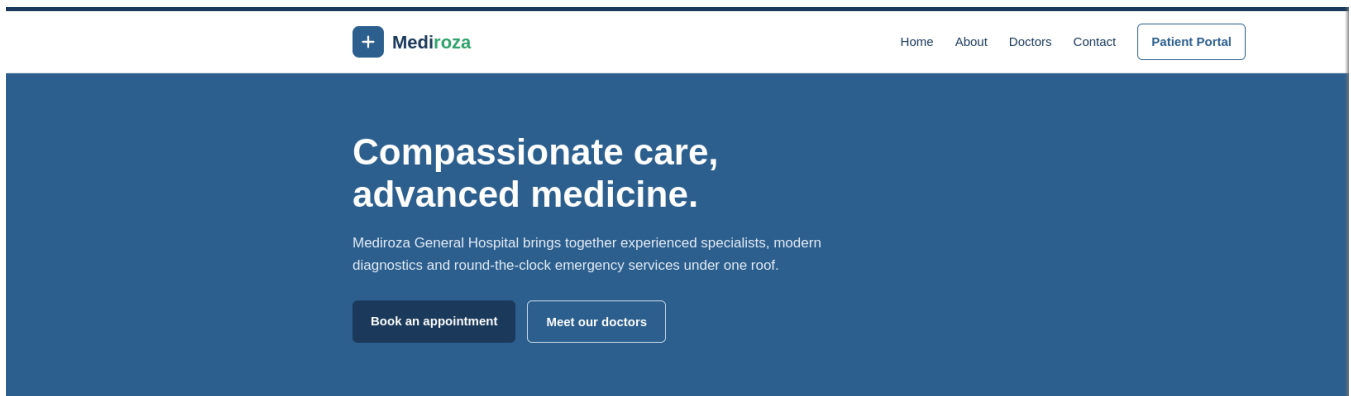
```

```

<section class="hero">
  <div class="container">
    <h1>Compassionate care, advanced medicine.</h1>
    <p>Mediroza General Hospital brings together experienced specialists, modern diagnostics and round-the-clock emergency services under one roof.</p>
    <div class="hero-actions">
      <a class="btn btn-primary" href="/contact.html">Book an appointment</a>
      <a class="btn btn-ghost" href="/doctors.html">Meet our doctors</a>
    </div>
  </div>
</section>

<section class="section">
  <div class="container">
    <h2>Our Departments</h2>
    <p class="lead">Comprehensive specialist care backed by a dedicated team and up-to-date medical technology.</p>
    <div class="grid">
      <div class="card">
        <span class="icon-tile"><svg viewBox="0 0 24 24"><path d="M12 21s-7-4.5-9-9a4.5 4.5 0 0 1 9-2 4.5 4.5 0 0 1 9 9 9 9 2"/></svg></span>
        <h3>Cardiology</h3>
        <p>Diagnosis and treatment of heart conditions with modern imaging and monitoring.</p>
      </div>
      <div class="card">
        <span class="icon-tile"><svg viewBox="0 0 24 24"><path d="M12 5v14M5 12h14"/></svg></span>
        <h3>Emergency & Trauma</h3>
        <p>A fully staffed 24/7 emergency unit ready for critical and urgent care.</p>
      </div>
      <div class="card">
        <span class="icon-tile"><svg viewBox="0 0 24 24"><circle cx="12" cy="8" r="4"/><path d="M4 21c0-4 4-6 8-6s 2 8 6"/></svg></span>
        <h3>Pediatrics</h3>
        <p>Gentle, specialised care for infants, children and adolescents.</p>
      </div>
      <div class="card">
        <span class="icon-tile"><svg viewBox="0 0 24 24"><rect x="4" y="4" width="16" height="16" rx="2" /><path d="M12 9v6M9 12h6"/></svg></span>
        <h3>Radiology</h3>
        <p>X-ray, ultrasound and CT imaging with fast, accurate reporting.</p>
      </div>
      <div class="card">
        <span class="icon-tile"><svg viewBox="0 0 24 24"><path d="M7 3h10M8 3v1-4 8a2 2 0 0 2 3h12a2 2 0 0 2 3l-4-8v3"/></svg></span>
        <h3>Pharmacy</h3>
        <p>On-site pharmacy stocking prescription and over-the-counter medicines.</p>
      </div>
      <div class="card">
        <span class="icon-tile"><svg viewBox="0 0 24 24"><path d="M3 12h4l2 6 4-14 2 8h6"/></svg></span>
        <h3>Diagnostics Lab</h3>
        <p>Accredited pathology laboratory for blood work and clinical testing.</p>
      </div>
    </div>
  </div>
</section>

```



Patient Portal

Sign in to view your lab reports and results.

Username

Password

☐ Show password

Sign in

Patient Portal

Sign in to view your lab reports and results.

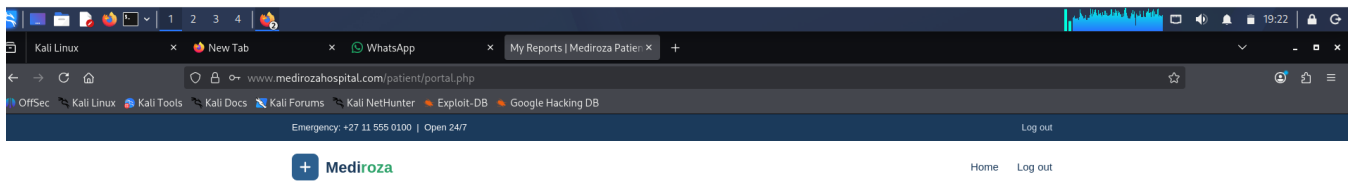
Username not found

Username

Password

☐ Show password

Sign in



My lab reports

Your reports are password protected. Use the password sent to you by the laboratory to open each file.

Pathology Report - S. Dlamini

Lab Ref LR-2024-1187 | 2024-11-04 | PDF (encrypted)

Download

Pathology Report - P. Reddy

Lab Ref LR-2024-1192 | 2024-11-05 | PDF (encrypted)

Download

Pathology Report - E. Thompson

Lab Ref LR-2024-1205 | 2024-11-06 | PDF (encrypted)

Download

My Report

INT - All

Password

1 new me

W3-PM1

PDF Hash

patient_r

C:\Users\adwak\Downloads\WK-4%20PDF\patient_report_1.pdf

1 of 1

Chat

+

Mediroza

General Hospital

14 Cedar Avenue, Johannesburg, South Africa
Tel: +27 11 555 0100
medirozahospital.com

PATHOLOGY LABORATORY REPORT

CONFIDENTIAL

Patient Name:

Sipho Dlamini

Patient ID:

MG-P-10231

Date of Birth:

1984-06-12

Gender:

Male

Report Date:

2024-11-04

Referring Dr:

Dr. Anita Naicker

Specimen:

Whole blood (EDTA)

Lab Ref No:

LR-2024-1187

TEST	RESULT	UNIT	REFERENCE RANGE	FLAG
Haemoglobin	13.9	g/dL	13.0 - 17.0	Normal
White Cell Count	11.8	x10 ⁹ /L	4.0 - 11.0	HIGH
Platelet Count	245	x10 ⁹ /L	150 - 400	Normal
Fasting Glucose	5.4	mmol/L	3.9 - 5.5	Normal
Creatinine	88	umol/L	64 - 104	Normal

Microsoft Store

+

Mediroza

General Hospital

14 Cedar Avenue, Johannesburg, South Africa
Tel: +27 11 555 0100
medirozahospital.com

PATHOLOGY LABORATORY REPORT

CONFIDENTIAL

Patient Name:

Priya Reddy

Patient ID:

MG-P-10244

Date of Birth:

1991-02-28

Gender:

Female

Report Date:

2024-11-05

Referring Dr:

Dr. Johan van der Merwe

Specimen:

Serum

Lab Ref No:

LR-2024-1192

TEST	RESULT	UNIT	REFERENCE RANGE	FLAG
Total Cholesterol	6.4	mmol/L	< 5.0	HIGH
LDL Cholesterol	4.1	mmol/L	< 3.0	HIGH
HDL Cholesterol	1.2	mmol/L	> 1.0	Normal
Triglycerides	1.8	mmol/L	< 1.7	HIGH
Fasting Glucose	4.9	mmol/L	3.9 - 5.5	Normal

+

Mediroza

General Hospital

14 Cedar Avenue, Johannesburg, South Africa
Tel: +27 11 555 0100
medirozahospital.com

PATHOLOGY LABORATORY REPORT

CONFIDENTIAL

Patient Name:

Emily Thompson

Patient ID:

MG-P-10258

Date of Birth:

1978-09-03

Gender:

Female

Report Date:

2024-11-06

Referring Dr:

Dr. Ahmed Kara

Specimen:

Serum

Lab Ref No:

LR-2024-1205

TEST	RESULT	UNIT	REFERENCE RANGE	FLAG
Haemoglobin	11.4	g/dL	12.0 - 15.5	LOW
Ferritin	9	ug/L	15 - 150	LOW
TSH	3.1	mIU/L	0.4 - 4.0	Normal
Vitamin D	42	nmol/L	50 - 125	LOW
Fasting Glucose	5.1	mmol/L	3.9 - 5.5	Normal

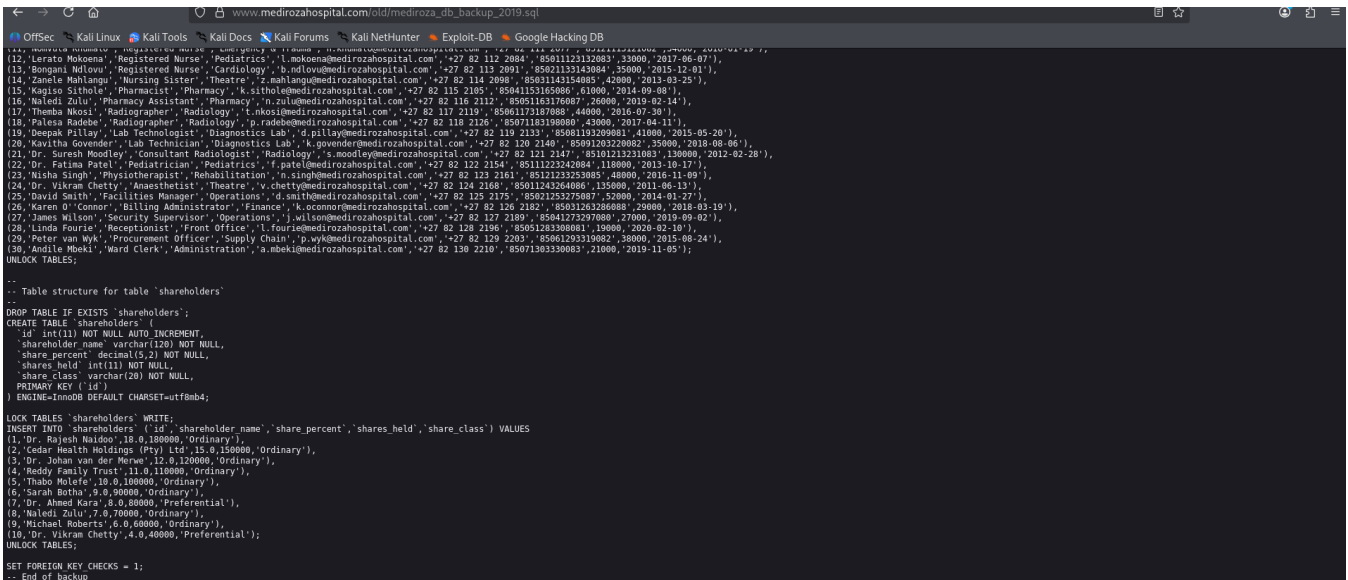
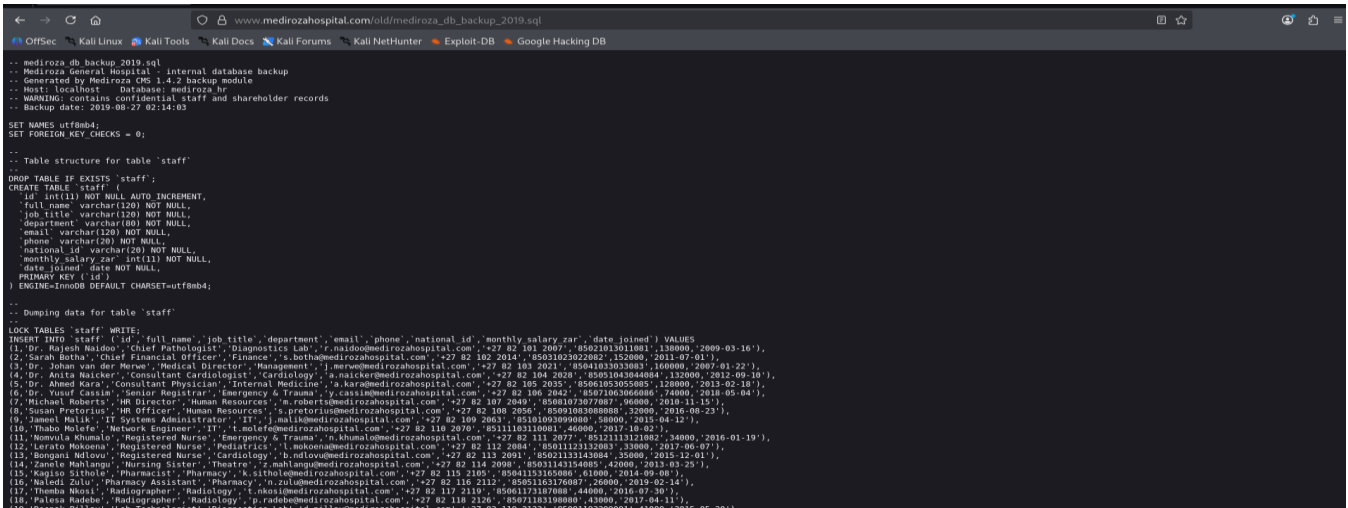
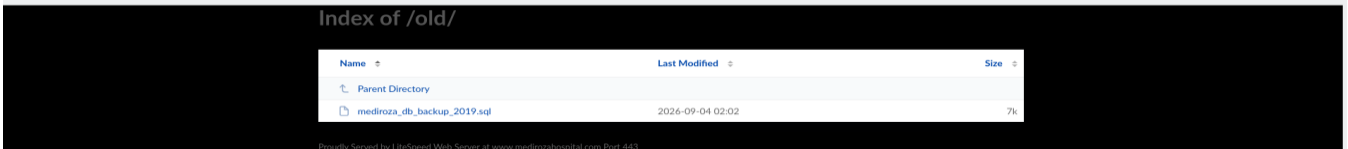
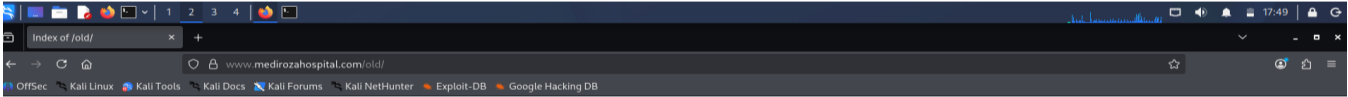
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```
(kali@kali)-[~]
└─$ curl https://www.medirozahospital.com/robots.txt

# robots.txt
User-agent: *
Disallow: /patient/
Disallow: /staff/
Disallow: /old/

Sitemap: https://medirozahospital.com/sitemap.xml

(kali@kali)-[~]
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08 — Conclusion

The Week 4 Mediroza engagement demonstrates how a black-box assessment can progress from basic reconnaissance to analysis of significant confidentiality risks. The public-facing application exposes a patient portal, while robots.txt reveals role-specific and legacy paths that help map the application's structure. The supplied evidence also confirms three encrypted pathology reports in the patient-report area, placing authentication and document security at the centre of the assessment.

The broader workflow examines whether authentication can be bypassed through unsafe input handling, whether protected PDFs can withstand offline password attacks, and whether legacy server resources expose database backups containing confidential employee or shareholder data. These stages illustrate how a seemingly minor weakness can become more significant when combined with weak access control, predictable document passwords, or poorly secured backup data.

The recommended remediation approach is layered: secure database queries, enforce authorization across sensitive resources, remove legacy content from public access, store backups outside the web root, strengthen document-password practices, minimize sensitive-data exposure, and monitor unusual authentication and file-access activity. A focused retest should follow remediation to confirm that identified attack paths have been closed.

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